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Boundary Layer on a Flat Plate with Radiation

CFD Investigation of Coupled Surface Radiation on Thermal Boundary Layer Development

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Subject: Boundary Layer Theory | Academic Project Presentation

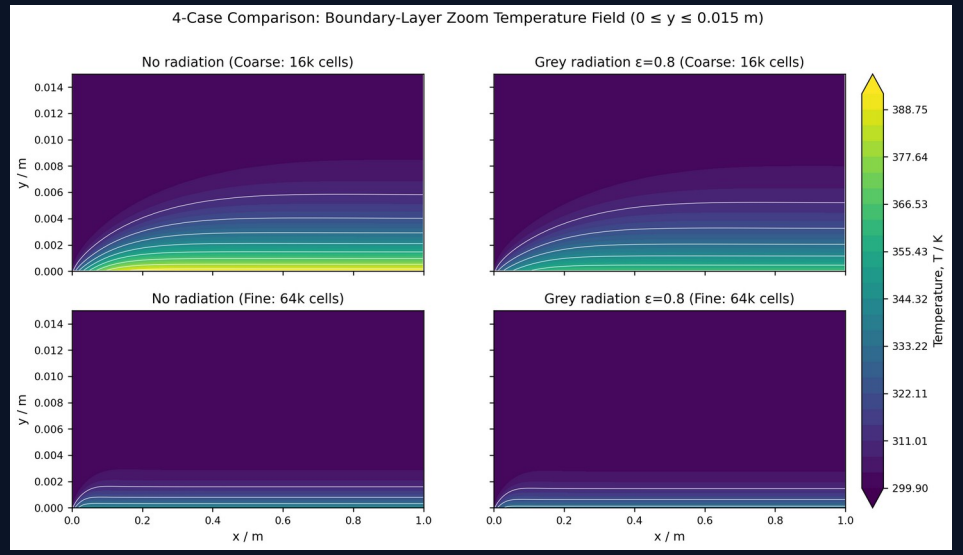


Fig. 1: Near-wall thermal boundary layer temperature fields across the 2 × 2 matrix.

Neglecting Surface Radiation Overpredicts Wall Temperatures

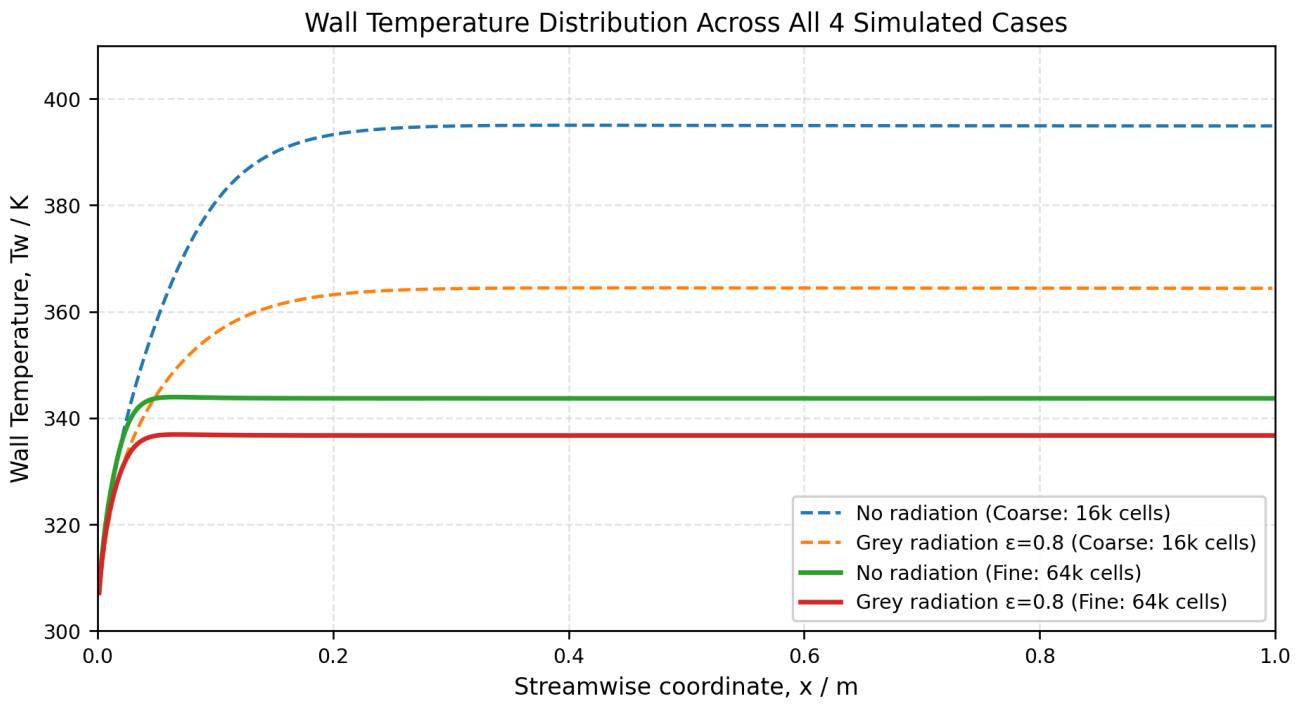


Fig. 2: Streamwise wall temperature distribution along heated flat plate.

Convective Cooling Decays Downstream

- Plate wall temperature depends strictly on competing heat rejection pathways.
- Convective cooling drops downstream as hydrodynamic boundary layers thicken.
- Standard CFD workflows frequently omit radiation, overpredicting component peak temperatures.

Heater Heat Flux: $q'' = 1000$ W/m²

Uniform electric heating over 1.0 m plate with T_ambient = 300 K, U_inf = 2.0 m/s

Newton-Raphson Solver Enforces Non-Linear Surface Heat Balance

Conjugate Energy Conservation at Wall Face

1. Surface Heat Balance (Enforced at Wall Face):

$$q''_{heater} = q''_{conv} + q''_{rad} = 1000\ W/m^2$$

2. Stefan-Boltzmann Quartic Radiation Emission:

$$q''_{rad} = \varepsilon\,\sigma\,(T_w^4 - T_{sur}^4)$$

where $\varepsilon = 0.80$, $\sigma = 5.670 \times 10^{-8}\ W/(m^2 \cdot K^4)$, $T_{sur} = 300\ K$

3. In-Situ Newton-Raphson Wall Root-Finding:

$$T_{w,new} = T_{w,old} - f(T_w) / f'(T_w)$$

$f(T_w) = -k\,(\partial T/\partial y)_w + \varepsilon\,\sigma\,(T_w^4 - T_{sur}^4) - q''_{heater} = 0$

Boundary Condition Mechanics

- Solves non-linear quartic radiative flux at every boundary face.
- Enforces local face heat balance by construction during each PIMPLE step.
- Newton-Raphson scheme converges to double precision within four iterations.

ε = 0.80

Grey-diffuse emissivity to 300 K external enclosure

< 4 Iter

Fast quadratic convergence per boundary face

Domain Mesh Resolution and 2 × 2 Simulation Matrix

2 × 2 Factorial Computational Matrix

Case Identifier	Grid (nx×ny)	Cells	Emissivity ε	Heater q″
noRadiation_coarse	200 × 80	16,000	0.0 (Pure)	1000 W/m²
noRadiation_fine	400 × 160	64,000	0.0 (Pure)	1000 W/m²
surfaceRadiation_coarse	200 × 80	16,000	0.8 (Coupled)	1000 W/m²
surfaceRadiation_fine	400 × 160	64,000	0.8 (Coupled)	1000 W/m²

- Domain includes 1.0 m unheated inflow section to develop boundary layer.
- Geometric wall-normal grading of 80:1 concentrates 41 cells in 5 mm.
- Fine 64,000-cell mesh maintains zero non-orthogonality and skewness.

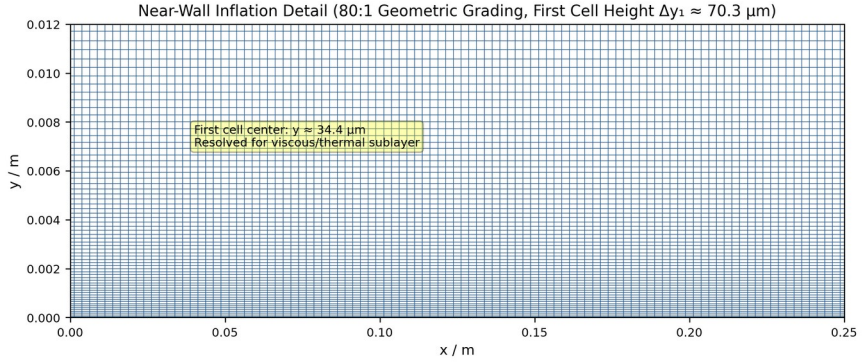


Fig. 3: Structured inflation mesh near heated plate leading edge (80:1 grading).

Skin Friction Validates Against Blasius Within 0.11%

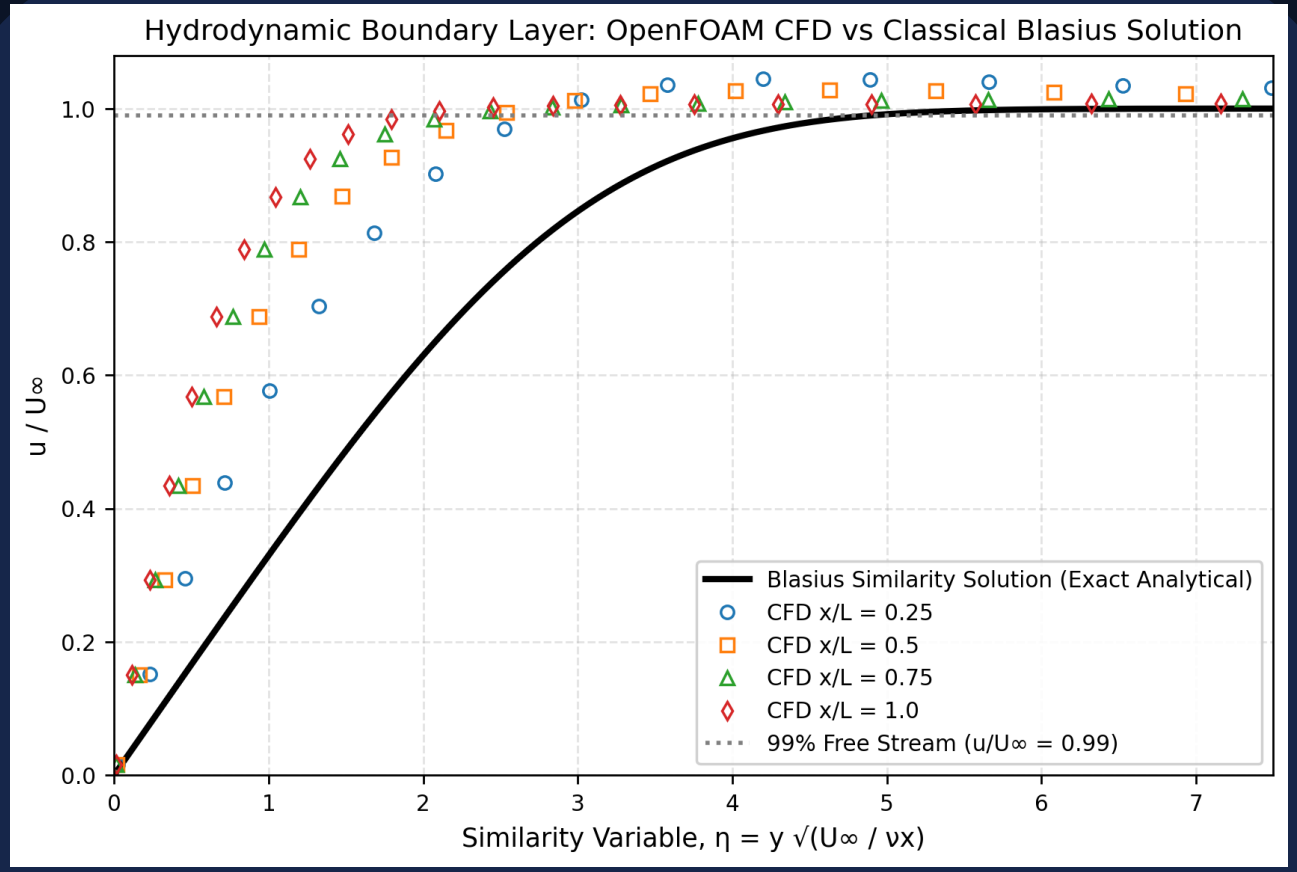


Fig. 4: Non-dimensional streamwise velocity collapse vs exact Blasius curve.

0.11%

Skin-friction relative error vs Blasius (Cf = 1.874e-3 vs 1.872e-3)

1.26×10^5

Plate Reynolds number (Re_L = U_inf L / ν) confirms laminar flow

Invariant

Velocity profiles strictly identical between radiating and baseline cases

Surface Radiation Depresses Peak Wall Temperature by 6.98 K

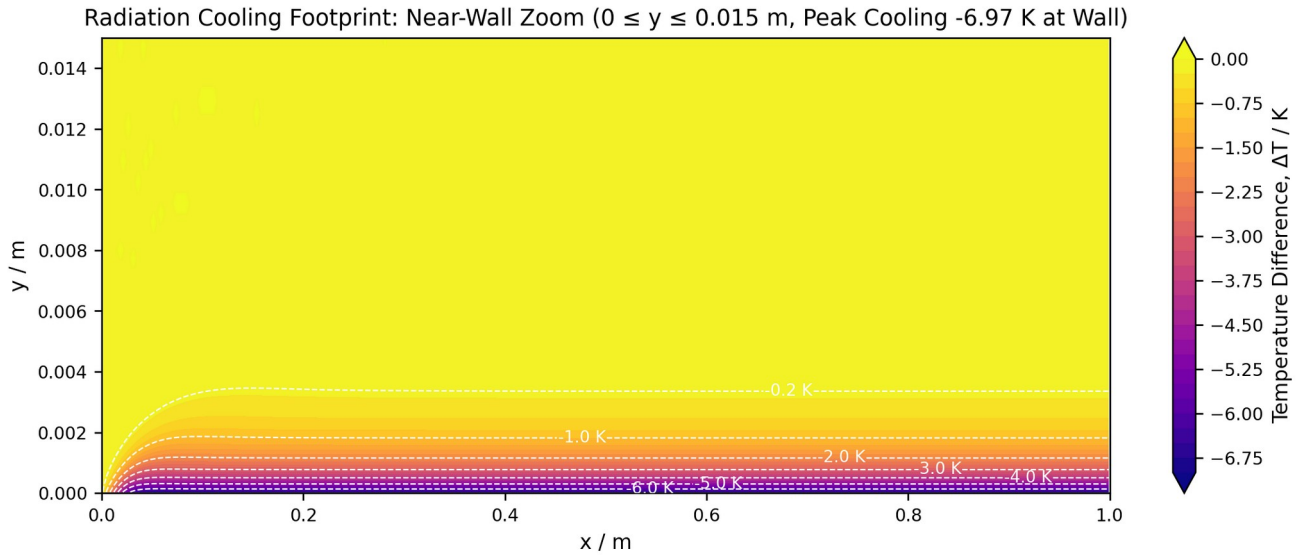


Fig. 5: Boundary layer temperature difference field (T_radiation - T_baseline).

6.98 K Drop

Peak Wall Temperature Reduction (Fine Grid)

- Radiation lowers plate wall temperature along the entire heated length.
- Cooling effect intensifies downstream, reaching maximum depression of 6.98 K.
- Thermal boundary layer thickness remains virtually unchanged across radiation cases.

Fine grid trailing edge: 336.70 K vs 343.68 K pure convection.

Surface Radiation Carries 21.3% of Total Heat Along Plate

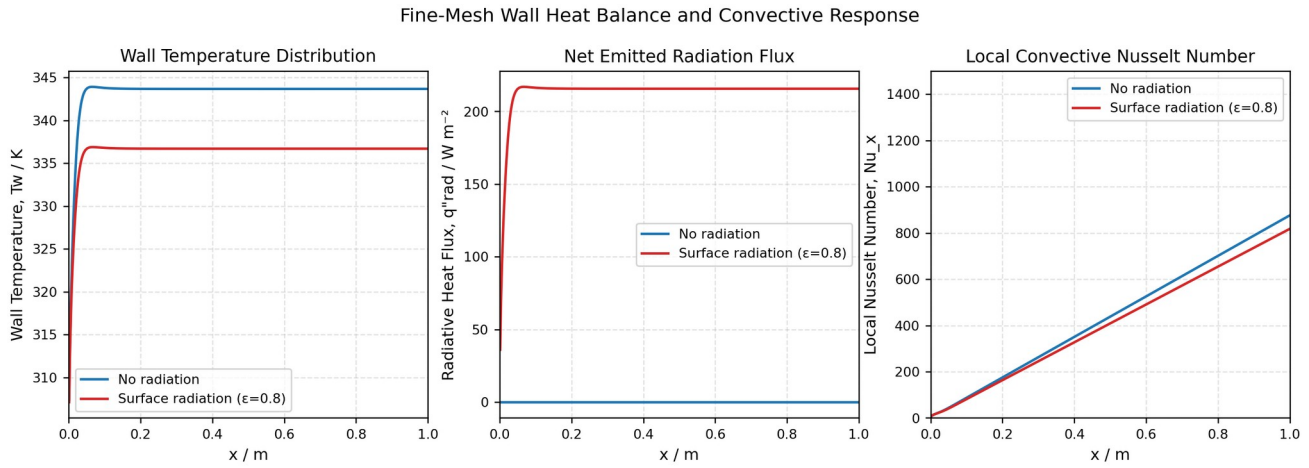


Fig. 6: Streamwise convective vs radiative heat-flux split and Nusselt number.

21.3%

Plate-Averaged Radiative Load

Mean flux: 213.2 W/m² (preliminary, fine-grid)

21.6%

Trailing-Edge Radiative Load

Trailing flux: 215.6 W/m² at x = 1.0 m

- Surface radiation relieves over one-fifth of the total heat dissipation load.
- Radiative heat flux remains nearly uniform along the entire plate surface.
- Convective flux drops downstream as thermal boundary layer growth increases resistance.

Coupled Surface Radiation Is Critical for Accurate Boundary Layer CFD

Objective Achieved: Successfully incorporated surface radiation and quantified its thermal boundary layer impact.

Key Findings & Numerical Insights

1. Surface Radiation Impact:
Relieves 21.56% of total heat flux and cools peak wall by 6.98 K (336.70 K vs 343.68 K).

2. Grid Sensitivity Disclosed:
Coarse (16k) overpredicts wall rise (394.9 K); fine grid (64k) resolves sublayer gradient.

3. Future Scope & Refinement:
Execute 3rd Richardson mesh for ASME GCI and incorporate variable air properties.

Student Credentials & References

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Key Formal References:

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